

Honors Analysis

First midterm

October 15th, 2025

Question 1. Write explicit examples of the following, whenever possible. One of the examples is not possible. *No justification is required.*

- (a) A sequence of real numbers that has two convergent subsequences with different limits.
- (b) A set that is connected but not path-connected.
- (c) A continuous bijective function $f : M \rightarrow N$, between two metric spaces M and N , such that the inverse function $f^{-1} : N \rightarrow M$ is not continuous.
- (d) A surjective continuous function $f : M \rightarrow N$ between two metric spaces M and N , so that N is connected but M is not.
- (e) A continuous function $f : [0, 1] \rightarrow \mathbb{R}$ and a closed subset $A \subset [0, 1]$ such that $f(A)$ is not closed.

Hint. For part (d), pay attention to which of the two spaces M and N is connected and which is not.

Question 2. Let A and B be two subsets of $\mathbb{R}^n$. We define their sum $A + B$ as

$$A + B := \{a + b : a \in A, b \in B\}.$$

- (a) If A and B are compact, prove that $A + B$ is compact.
- (b) If A is compact and B is closed, prove that $A + B$ is closed.
- (c) Give an example of two closed sets A and B such that $A + B$ is not closed.

Hint. For part (c), I think it is intuitively easier to first find an example for $A - B$. An example for $A + B$ can be constructed by replacing B with $-B$. Consider finding sets A and B such that 0 is a boundary point of $A - B$ but $0 \notin A - B$.